Dataset 1: PJM Hourly Load Dataset

Each row in the PJM Hourly Load dataset represents a single metered electricity load observation for a specific load area within the PJM Interconnection grid. Each observation corresponds to one hour of electricity demand recorded at one load area at a specific point in time.

Column Name	Description	Example Values
datetime_beginning_utc	Timestamp of the electricity demand observation in Coordinated Universal Time	1/1/2025 5:00:00 AM
datetime_beginning_ept	Timestamp of the electricity demand observation in Eastern Prevailing Time	1/1/2025 12:00:00 AM
nerc_region	North American Electric Reliability Corporation region for the observation	RFC
mkt_region	PJM market region for the observation	MIDATL, WEST
zone	Transmission zone identifier	DOM, AEP, RTO
load_area	Specific load area within the zone	DOM, AECO, RECO
mw	Metered electricity demand for the load area measured in megawatts	12464.698, 870.261
is_verified	Indicator of whether the observation has been verified by the data provider	True, False

Dataset 2: NOAA Daily Weather Summary Dataset

Each row in the NOAA Daily Weather Summary dataset represents a single day of weather observations recorded at one weather station in the Charlottesville, Virginia area. Each observation corresponds to one station-date pair, including daily precipitation totals and temperature extremes.

Column Name	Description	Example Values

STATION	Unique identifier for the weather station	USW00003759, USC00443213
NAME	Name and location of the weather station	CHARLOTTESVILLE 2 SSE, VA US
DATE	Date of the weather observation	2025-01-01
PRCP	Total precipitation recorded for the day in inches	0.00, 0.21, 0.63
TMIN	Minimum temperature recorded for the day in degrees Fahrenheit	22.0, 33.0
TMAX	Maximum temperature recorded for the day in degrees Fahrenheit	35.0, 50.0

Dataset 3: Final Analytical Dataset

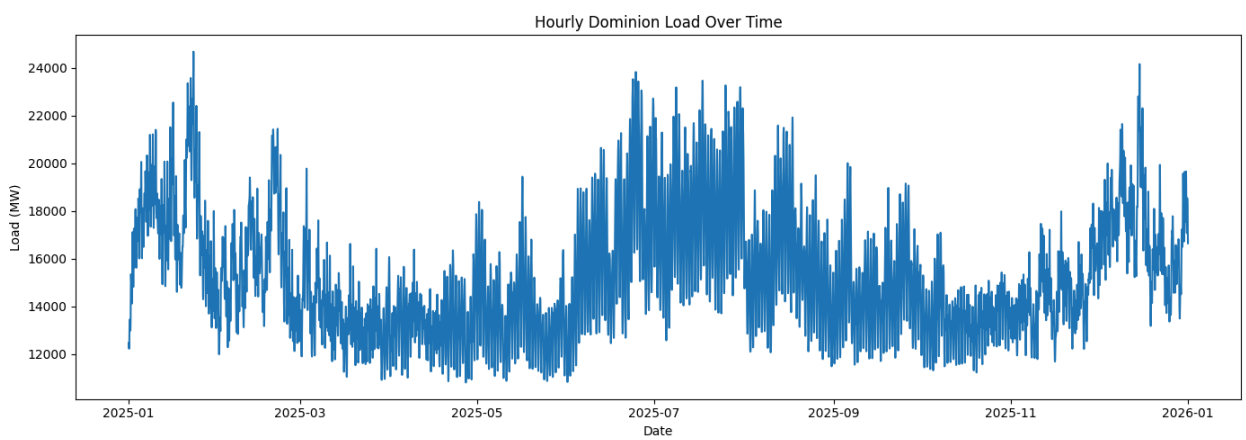
Each row in the final analytical dataset represents a single hourly electricity demand observation for the Dominion (DOM) load area, merged with corresponding daily weather data for the Charlottesville, Virginia region and processed for time-series forecasting. Each observation corresponds to one hour of metered load paired with the daily weather conditions on that date, along with a calendar indicator variable. This dataset reflects the cleaned and transformed data used for all ARIMA and ARIMAX model training and evaluation in the study.

Column Name	**Description**	**Example Values**
beginning_est_in_datetime	Timestamp of the electricity demand observation in Eastern Standard Time; serves as the time-series index	2025-01-01 00:00:00-05:00
mw	Electricity demand for the Dominion load area measured in megawatts; serves as the forecast target variable	12464.698, 16641.733
PRCP	Average daily precipitation across selected Charlottesville-area weather stations in inches; used as an exogenous predictor in the ARIMAX model	0.207, 0.000
TMIN	Average daily minimum temperature across selected	33.67, 23.00

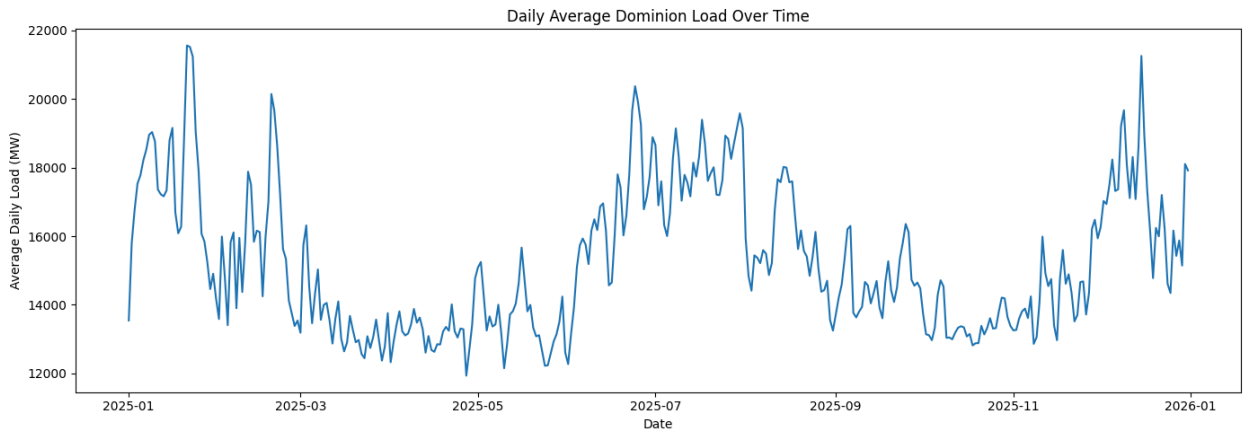
	weather stations in degrees Fahrenheit; used as an exogenous predictor in the ARIMAX model	
TMAX	Average daily maximum temperature across selected weather stations in degrees Fahrenheit; used as an exogenous predictor in the ARIMAX model	52.00, 37.67
is_weekend	Binary indicator of whether the observation falls on a Saturday or Sunday; used as a calendar predictor in the ARIMAX model	0 (weekday), 1 (weekend)

Exploratory Visualizations:
The following exploratory plots were created to establish familiarity with the dataset:

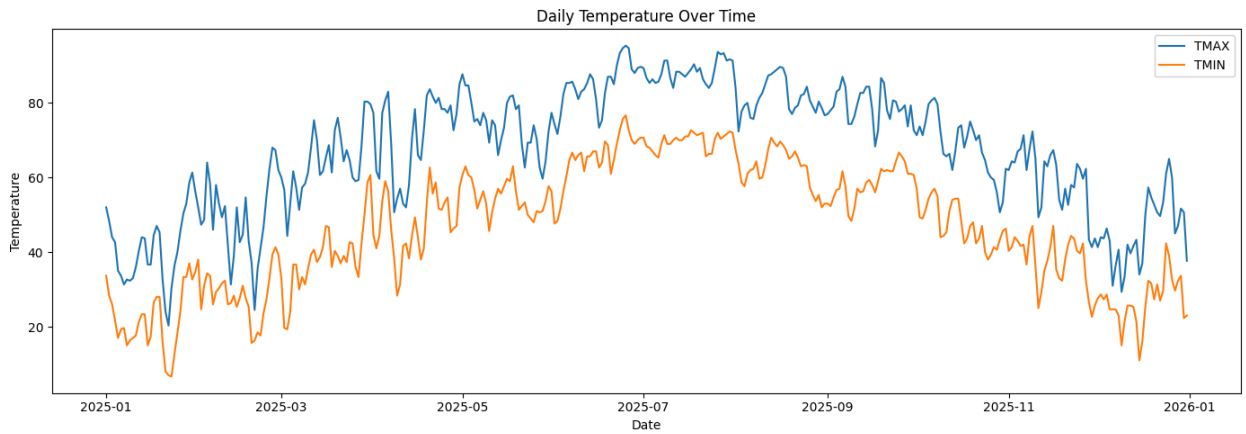
Hourly Load Over Time



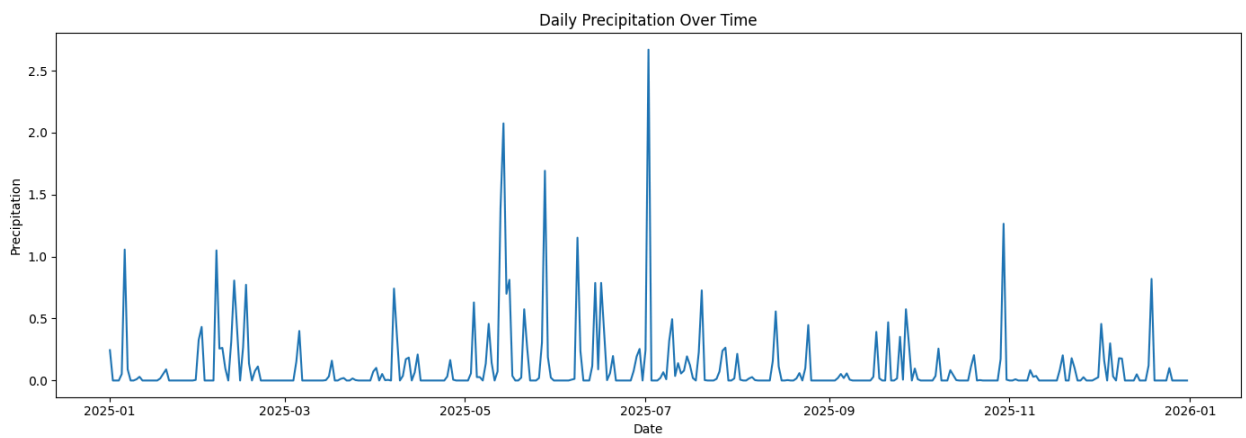
Daily Load Over Time



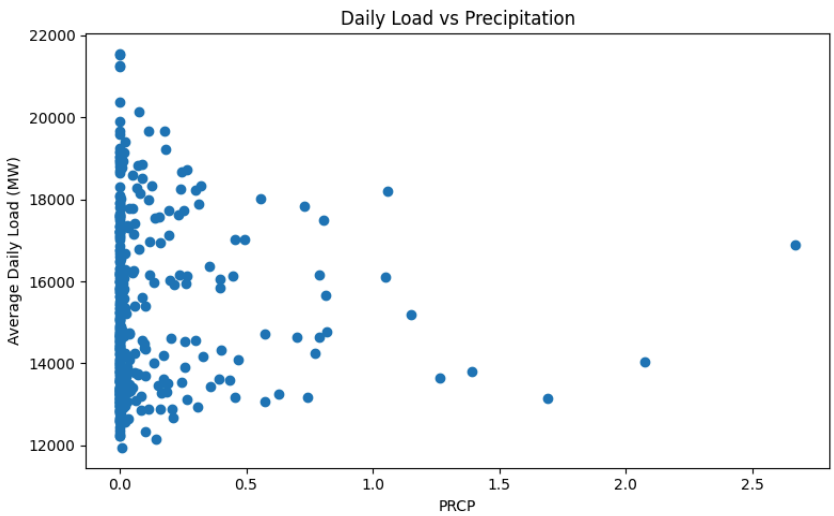
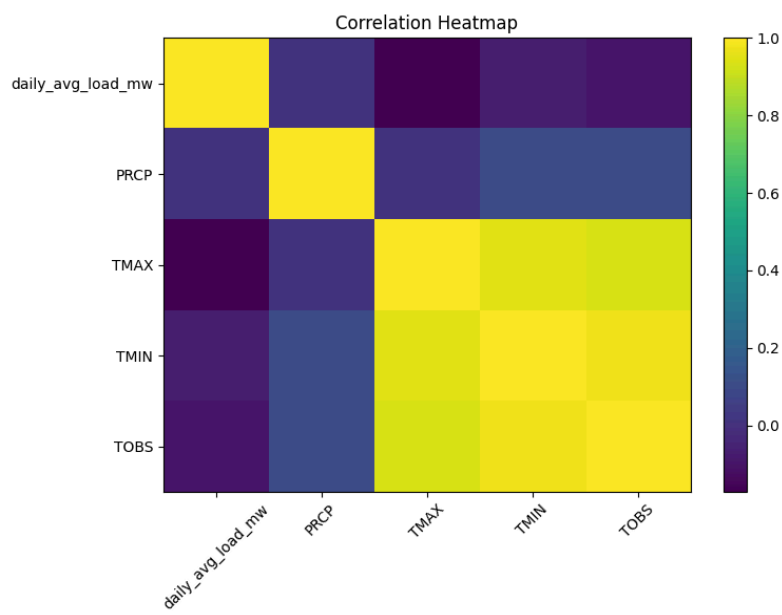
Daily Temperature Mins and Maxes Over Time



Daily Precipitation Over Time



Daily Load Does Not Correlate With Precipitation



Correlation Heatmap of Forecasting Variables